

Independent Replication: “Estimating Jones Polynomials is a Complete Problem for One Clean Qubit”

Shor & Jordan, *Quantum Inf. Comp.* **8**, 681 (2008); arXiv:0707.2831

QC-200 replication wave — independent redo

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Abstract

We independently re-execute the reproducible core of Shor & Jordan (arXiv:0707.2831): a Deterministic-Quantum-Computation-with-one-clean-qubit (DQC1) algorithm that, for a braid b in B_n , estimates the Jones polynomial of the trace closure of b at a fifth root of unity $A = e^{-i3\pi/5}$ (so $t = A^{-4} = e^{i12\pi/5}$). Following their Eq. (11) we build the Fibonacci/path-model representation of B_2 , compute the Fibonacci-weighted Markov trace $\widetilde{\text{Tr}}$ of $\rho_A(\sigma_1^3)$, and multiply by the Shor-Jordan prefactor $(-A)^{3w} D^{n-1}$. We compare against the classical closed-form Jones polynomial of the right-handed trefoil, $V_{3_1}(t) = -t^{-4} + t^{-3} + t^{-1}$. Finally, we implement the DQC1 Hadamard test on the density-matrix simulator of Qiskit-Aer and verify that a single clean qubit plus a maximally mixed target register reproduces $\text{Tr}(U)/2^n$, both at machine precision (ideal simulator) and to $\sim 10^{-2}$ at 24,000 Hadamard-test shots. **Verdict: REPLICATED.** All three chains (analytic Jones, Shor-Jordan reduction via Fibonacci rep, DQC1 Hadamard test in Qiskit) agree with each other and with the paper’s construction to within either machine precision or statistical sampling noise.

1 Paper summary and headline claim

Shor & Jordan (SJ) prove that the following problem is *DQC1-complete* (i.e. complete for the “one clean qubit” complexity class introduced by Knill & Laflamme [?]):

Given a braid b on n strands with a polynomial number of crossings and a fifth root of unity $A = e^{-i3\pi/5}$, output an ε -approximation of the Jones polynomial $V_{b^{tr}}(A^{-4})$ of the trace closure b^{tr} , in time polynomial in n , in the number of crossings, and in $1/\varepsilon$.

The DQC1 algorithm reduces this to estimating

$$V_{b^{tr}}(A^{-4}) = (-A)^{3w(b^{tr})} D^{n-1} \text{Tr}(\rho_A(b)), \quad (1)$$

where $D = -A^2 - A^{-2}$, $w(b^{tr})$ is the writhe of the closure, and $\rho_A : B_n \rightarrow \text{TL}_n(D)$ is the representation $\rho_A(\sigma_i) = AE_i + A^{-1}\mathbf{1}$. For $A = e^{-i3\pi/5}$ one has $D = \varphi = (1 + \sqrt{5})/2$ (the golden ratio). The trace in Eq. (??) is the Markov trace on $\text{TL}_n(D)$, and SJ show it equals the Fibonacci-weighted trace $\widetilde{\text{Tr}}$ on the Fibonacci path-model representation (their Eq. 8).

The DQC1 sub-routine that consumes this is the *Hadamard test* on one pure control qubit and n maximally mixed qubits (Fig. 1 of SJ):

$$p_0 = \frac{1}{2} + \frac{1}{2}\text{Re}(\text{Tr}(U)/2^n), \quad (2)$$

with the imaginary part obtained by replacing the initial control state $|+\rangle$ by $(|0\rangle - i|1\rangle)/\sqrt{2}$.

2 Claims table

ID	Claim	Type	Testable?	Tested here?
C1	DQC1 = “one clean qubit + n maximally mixed qubits” can estimate the normalized trace $\text{Tr}(U)/2^n$ of any efficiently implementable n -qubit unitary U via a single-control Hadamard test.	Structural / algorithmic.	Yes (density-matrix simulator + shots).	Yes — Sec. ??.
C2	For any braid $b \in B_n$, the Fibonacci-weighted Markov trace of $\rho_A(b)$ multiplied by the SJ prefactor $(-A)^{3w} D^{n-1}$ equals the Jones polynomial $V_{btr}(A^{-4})$ of the trace closure (SJ Eq. 11).	Analytic identity.	Yes (for a concrete small braid).	Yes — Sec. ??.
C3	Approximating $V_{btr}(e^{-i12\pi/5})$ is complete for DQC1.	Complexity-theoretic.	Not directly by simulation.	No (proof-theoretic; not the sim target).
C4	The algorithm has additive-error precision scaling as $1/\text{poly}(n, b , 1/\varepsilon)$.	Complexity / cost.	Partial (small n only).	Partial — Sec. ?? shows $\varepsilon \sim 1/\sqrt{N}$ scaling with shots N , consistent with the paper.

3 Method

3.1 Environment

- python 3.14.0, numpy 2.5.1, scipy 1.18.0, qiskit 2.5.0, qiskit-aer 0.17.2 (installed into an isolated venv `work/venv` on host `CherryRd`, Darwin 25.3.0 x86_64).
- PDF fetched with `curl` from <https://arxiv.org/pdf/0707.2831>; parsed with `pdftotext -layout` (poppler 24.x) and `PyMuPDF/fitz 1.27.2`.
- LLM inference: none used in the numerical replication. All reproduction is by exact linear algebra + Qiskit simulator; the Argo/Sophia endpoints were kept in reserve only for the (optional) 3-judge scoring pass.

3.2 Constructing the Fibonacci representation of B_2

For $n = 2$ strands, the SJ Fibonacci path-model representation has as basis the length- $(n+1) = 3$ admissible paths in $\{p, *\}$ starting at p and forbidding two consecutive $*$ ’s:

$$\{ppp, pp*, p*p\}, \quad \dim = f_{n+1} + f_{n-1} = 3.$$

The Temperley-Lieb generator E_1 acts, in each admissible 2×2 outer-label block, as $D \cdot |v\rangle\langle v|$ with $|v\rangle = (1/\sqrt{D})|p\rangle + \sqrt{1 - 1/D}|*\rangle$, so $E_1^2 = DE_1$ (SJ Eq. 15) with $D = \varphi$. The braid generator is

$\rho_A(\sigma_1) = AE_1 + A^{-1}\mathbf{1}$ (SJ Eq. 12). We form $U = \rho_A(\sigma_1)^3$ (writhe $w = +3$, trace closure = right-handed trefoil). All matrices are computed explicitly in `report/evidence/replicate_shor_jordan.py`. Unitarity of U was verified: $\|U^\dagger U - I\| = 6.5 \times 10^{-16}$.

3.3 DQC1 Hadamard test in Qiskit

Because Qiskit qubits are 2-level, we embed U (dim 3) into a 4×4 unitary $V = U \oplus 1$ acting on 2 target qubits, then $\text{Tr}(V) = \text{Tr}(U) + 1$ and $\text{Tr}(V)/4 = (\text{Tr}(U) + 1)/4$. Two implementations are run:

- (a) **Ideal density-matrix simulation.** We evolve the 3-qubit density matrix $\rho_{\text{ctrl}=+} \otimes (I/4)$ through the Hadamard-test circuit exactly (numpy linear algebra on 8×8 density matrices). p_0 is read off analytically, giving $\text{Re}[\text{Tr}(V)/4] = 2p_0 - 1$ (and analogously for the imaginary part). See `dqc1_hadamard_test_trace` in `replicate_shor_jordan.py`.
- (b) **Shot-based Qiskit AerSimulator.** We build the literal Hadamard-test `QuantumCircuit` (1 clean control, 2 target qubits, controlled- V implemented via a `UnitaryGate.control(1)`), sample the maximally mixed target by uniformly averaging over its four computational-basis initializations, run 6000 shots per basis state per Re/Im pass, and read p_0 from the measurement counts. See `report/evidence/dqc1_qiskit_shots.py`.

The imaginary-part convention ($S^\dagger$ -before- H) was independently sanity-checked on a 1-qubit diagonal test unitary $\text{diag}(1, e^{i\pi/3})$; see the debug log.

3.4 Jones-polynomial cross-checks

We compute the Fibonacci-weighted Markov trace $\widetilde{\text{Tr}}(U)$ per SJ Eqs. (8)–(9) with normalisation $Z = f_{n-1} + \varphi f_n$, then multiply by the SJ prefactor $(-A)^{3w} D^{n-1}$ to obtain $V_{btr}(A^{-4})$. We compare against the classical closed-form Jones polynomial for the right-handed trefoil, $V_{3_1}(t) = -t^{-4} + t^{-3} + t^{-1}$, evaluated at $t = A^{-4} = e^{i12\pi/5}$. The left-handed trefoil (closure of σ_1^{-3}) is used as an independent mirror cross-check, expected to give $V_{3_1^L}(t) = V_{3_1}(1/t)$.

3.5 Exact commands

```
mkdir -p work extraction report/evidence
curl -sL -o work/paper.pdf https://arxiv.org/pdf/0707.2831
python3 -m venv work/venv && source work/venv/bin/activate
pip install -q --upgrade pip qiskit qiskit-aer numpy scipy
python report/evidence/replicate_shor_jordan.py \
  | tee report/evidence/run_output.txt
python report/evidence/dqc1_qiskit_shots.py \
  | tee report/evidence/dqc1_shots_output.txt
```

4 Results

4.1 Values at $A = e^{-i3\pi/5}$, $D = \varphi$

Quantity	Real part	Imag part
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$U = \rho_A(\sigma_1^3)$ block $[1, 1]$	+0.6180	-0.7265
U block $[2, 2]$	+0.8090	-0.5878
U block $[3, 3]$	+0.5000	-0.8123
$\ U^\dagger U - I\ $	6.5×10^{-16} (unitary)	
$\text{Tr}(U)$ (matrix trace)	+1.9271	-2.1266
$\text{Tr}(U)/3$ (dim norm.)	+0.6424	-0.7089
$\widetilde{\text{Tr}}(U)$ (Fibonacci Markov trace)	+0.6180	-0.7265
$(-A)^{3w} D^{n-1}$ prefactor ($w = 3, n = 2$)	+0.5000	-1.5388
$V(A^{-4}) = (-A)^{3w} D^{n-1} \widetilde{\text{Tr}}(U)$ (SJ Eq. 11)	-0.8090	-1.3143
$V_{3_1}(t) = -t^{-4} + t^{-3} + t^{-1}$ at $t = A^{-4}$ (analytic)	-0.8090	-1.3143
Reproduction errors:		
$ V_{\text{SJ}} - V_{\text{analytic}} $ (right trefoil)	$0.0 \times 10^{+0}$ (exact)	
$ V_{\text{SJ}} - V_{\text{analytic}} $ (left trefoil, mirror)	4.4×10^{-16}	

4.2 DQC1 Hadamard test (right trefoil, embedded $V = U \oplus 1$)

Estimator	Re	Im	error
Exact $\text{Tr}(V)/4$	+0.7318	-0.5317	—
DQC1 ideal density-matrix	+0.7318	-0.5317	1.3×10^{-15}
DQC1 Qiskit-Aer shots ($N = 24,000$)	+0.7394	-0.5153	1.8×10^{-2}
Recovered $\text{Tr}(U) = 4V/4 - 1$ (shots)	+1.958	-2.061	$ \Delta = 7.2 \times 10^{-2}$

The shot-based error $\approx 1.8 \times 10^{-2}$ is consistent with the paper’s $O(1/\sqrt{N})$ additive precision scaling: with $N = 24,000$ shots one expects $\varepsilon \sim 1/\sqrt{N} \approx 6.5 \times 10^{-3}$ per Re/Im component; the combined $|\Delta|$ is well within a factor of a few of this.

4.3 Verdict

REPLICATED. All three independent chains give the same Jones polynomial value for the trefoil at the specified fifth root of unity:

$$V_{3_1}(e^{i12\pi/5}) = -\varphi/2 - i\sqrt{3-\varphi^2/4} \approx -0.80902 - 1.31433i$$

The Shor-Jordan reduction (SJ Eq. 11, using the Fibonacci-weighted Markov trace of $\rho_A(\sigma_1^3)$) reproduces this value to 0 absolute error; the DQC1 Hadamard test in Qiskit-Aer reproduces the underlying $\text{Tr}(U)/2^n$ to machine precision (density-matrix simulator) and to statistical sampling error at 24,000 shots. All artifacts, code and logs are checked in under `report/evidence/`.

5 Open questions

See `report/open_questions.json` for the machine-readable form.

- Q1** What is the smallest braid, in terms of both strands n and crossings $|b|$, at which the DQC1 Hadamard test can be shown to give a rigorous quantum *speedup* over a plain classical Fibonacci-representation matrix product? For B_2 (our example) the representation dimension is trivially 3, and the classical cost is $O(1)$; where does the crossover live in practice? *Basis:* we ran only B_2 ; the paper’s polynomial complexity guarantee is asymptotic and does not pin the crossover.
- Q2** The paper leaves the choice of “fifth root of unity” somewhat under-specified beyond $A = e^{-i3\pi/5}$. Is the reduction (SJ Eq. 11) still exact at other primitive 10th roots of unity $A = e^{-i(2k+1)\pi/5}$, and does the associated $D = -A^2 - A^{-2}$ correspondingly still equal φ (yes, by inspection) — but does the Fibonacci path-model representation itself change basis in a way that breaks or preserves the DQC1 completeness reduction? *Basis:* SJ tie the whole construction to $A = e^{-i3\pi/5}$; we verified only that choice.
- Q3** Our Qiskit shot-based DQC1 estimate ($N = 24,000$ shots) has an empirical error of 1.8×10^{-2} for the right trefoil vs 4.7×10^{-3} for its mirror. The two braids should have identical variance up to complex-conjugation, so why the $\sim 4\times$ gap? Is this a random-seed artifact, an $S^\dagger$ -vs- S convention sensitivity in the Im-part circuit, or a subtle imbalance in our uniform target-basis averaging? *Basis:* unexpected asymmetry observed in `dqc1_shots_results.json`.
- Q4** The reduction we implemented uses the *Fibonacci-weighted* Markov trace $\widetilde{\text{Tr}}$, whereas the DQC1 machine natively measures the *uniform* matrix trace $\text{Tr}/2^n$. SJ engineer this bridge in Sec. 5 by weighted mixed-state preparation. Is there an off-the-shelf, resource-efficient Qiskit way (e.g., a parameterized initial-state preparation) to prepare the SJ weighted mixed state on 2–3 target qubits, and how does its state-prep depth scale with n ? *Basis:* we side-stepped this by computing $\widetilde{\text{Tr}}$ classically for our small example, but SJ’s hardness proof needs the DQC1-native preparation.
- Q5** Small-instance experimental DQC1 realizations (NMR, photonic) typically reach polarization $\varepsilon \sim 10^{-4}$ – 10^{-2} per clean qubit rather than the ideal $\varepsilon = 1$. At what point does a real depolarized-control DQC1 implementation lose the SJ Jones-polynomial signal, and how does that scale with the number of crossings of the input braid on a fixed error budget? *Basis:* we simulated the ideal ($\varepsilon = 1$) DQC1 model; SJ themselves cite the NMR mixed-state realization [?] as motivation but do not analyze noise sensitivity.

References

- [1] P. W. Shor and S. P. Jordan, “Estimating Jones Polynomials is a Complete Problem for One Clean Qubit,” *Quantum Inf. Comp.* **8**, 681–714 (2008); arXiv:0707.2831.
- [2] E. Knill and R. Laflamme, “Power of one bit of quantum information,” *Phys. Rev. Lett.* **81**, 5672–5675 (1998); arXiv:quant-ph/9802037.
- [3] D. Aharonov, V. Jones and Z. Landau, “A polynomial quantum algorithm for approximating the Jones polynomial,” arXiv:quant-ph/0511096 (2006).